

Figure 1

Contents

1	Packages, data and functions	1
1.1	Packages	1
1.2	Data	1
1.3	Functions	1
2	Figure 1 & Table 1	2
2.1	Models	2
2.2	Figure	3
2.3	Table	7
3	Versions of R and of the R-packages used	10

1 Packages, data and functions

1.1 Packages

```
library(tidyverse)
library(patchwork)
library(ggstance)
library(here)
library(lme4)
library(lmerTest)
library(kableExtra)
```

1.2 Data

```
load(here("Data/3_ForecastsResults/3_MergedResults/complete.forecasts.RData"))
```

1.3 Functions

```
conf_interval <- function(model, nd, response, renameResponse, cmult=1.96){
  nd[,response] <- predict(model, nd, re.form=NA)
  mm <- model.matrix(terms(model),nd)
  pvar1 <- diag(mm %*% tcrossprod(vcov(model),mm))
}
```

```

nd <- nd %>%
  dplyr::mutate(lower=!!as.name(response)-cmult*sqrt(pvar1),
                upper=!!as.name(response)+cmult*sqrt(pvar1),
                Response=renameResponse) %>%
  dplyr::rename(fit=!!as.name(response))
return(nd)
}

```

2 Figure 1 & Table 1

2.1 Models

```

## taxa
formula <-
  as.formula(
    RMSE ~ median_richness_center * light_treatment +
      (1 | Target) + (1 | bottle) + (1 | incubator)
  )

m.simplex <- lmer(
  formula,
  complete.forecasts %>%
    dplyr::filter(
      forecast.method == "Simplex EDM",
      Target != "Total_biomass",
      Target != "Mean_oxygen"
    )
)

## biomass
formula <-
  as.formula(RMSE ~ median_richness_center * light_treatment +
    (1 | composition) + (1 | incubator))

m.simplex.bio <- lmer(
  formula,

```

```
complete.forecasts %>%
  dplyr::filter(forecast.method == "Simplex EDM",
                Target == "Total_biomass")
)
```

2.2 Figure

```
dd.m.simplex <- complete.forecasts %>%
  dplyr::filter(forecast.method == "Simplex EDM",
                Target != "Total_biomass",
                Target != "Mean_oxygen")

newdat.m.simplex <-
  expand.grid(
    median_richness_center = seq(
      min(dd.m.simplex$median_richness_center),
      max(dd.m.simplex$median_richness_center),
      length.out = 100
    ),
    light_treatment = unique(dd.m.simplex$light_treatment)
  )

newdat.m.simplex <-
  conf_interval(m.simplex, newdat.m.simplex, "RMSE", "Forecast error (RMSE)")

nbot <- complete.forecasts %>%
  dplyr::filter(forecast.method == "Simplex EDM") %>%
  group_by(median_richness_center, light_treatment, bottle) %>%
  summarize(nspecies = n()) %>%
  group_by(median_richness_center, light_treatment) %>%
  summarize(nbottles = paste0("n=", n()))

figure1a <- complete.forecasts %>%
  dplyr::filter(Target != "Total_biomass",
                Target != "Mean_oxygen",
                forecast.method == "Simplex EDM") %>%
```

```

ggplot(aes(median_richness_center, RMSE, col = light_treatment)) +
geom_ribbon(
  data = newdat.m.simplex,
  aes(
    ymin = lower,
    ymax = upper,
    y = fit,
    group = light_treatment
  ),
  col = NA,
  fill = "grey",
  alpha = 0.3
) +
geom_line(data = newdat.m.simplex,
  size = 1.3,
  aes(y = fit, group = light_treatment)) +
geom_point(size = 2.5, alpha = 0.6) +
labs(x = "Centered median realized species richness",
  y = "Taxa abundance forecast error (RMSE)",
  col = "Light conditions") +
scale_color_manual(values = c("#C4961A", "#293352"),
  aesthetics = c("color", "fill")) +
theme_bw() +
geom_text(
  data = nbot,
  aes(x = median_richness_center, y = 3, label = nbottles),
  position = position_dodgev(height = 0.2),
  fontface = "bold",
  show.legend = F,
  # position = position_dodge(width = 0.5),
  size = 3.5
) +
ylim(0, 3.05)

#-----
#### Fig1b: Biomass forecast simplex

```

```

#-----

dd.m.simplex <- complete.forecasts %>%
  dplyr::filter(forecast.method == "Simplex EDM",
                Target == "Total_biomass")

newdat.m.simplex.bio <-
  expand.grid(
    median_richness_center = seq(
      min(dd.m.simplex$median_richness_center),
      max(dd.m.simplex$median_richness_center),
      length.out = 100
    ),
    light_treatment = unique(dd.m.simplex$light_treatment)
  )

newdat.m.simplex.bio <-
  conf_interval(m.simplex.bio,
                newdat.m.simplex.bio,
                "RMSE",
                "Forecast error (RMSE)")

figure1b <- complete.forecasts %>%
  dplyr::filter(Target == "Total_biomass", forecast.method == "Simplex EDM") %>%
  ggplot(aes(median_richness_center, RMSE, col = light_treatment)) +
  geom_ribbon(
    data = newdat.m.simplex.bio,
    aes(
      ymin = lower,
      ymax = upper,
      y = fit,
      group = light_treatment
    ),
    col = NA,
    fill = "grey",
    alpha = 0.3
  )

```

```

) +
geom_line(data = newdat.m.simplex.bio,
          size = 1.3,
          aes(y = fit, group = light_treatment)) +
geom_point(size = 2.5, alpha = 0.6) +
labs(x = "Centered median realized species richness",
     y = "Community biomass forecast error (RMSE)",
     col = "Light conditions") +
scale_color_manual(values = c("#C4961A", "#293352"),
                  aesthetics = c("color", "fill")) +
theme_bw() +
ylim(0, 3.05)

figure1a + figure1b +
plot_layout(guides = "collect") + plot_annotation(tag_levels = "a")

```

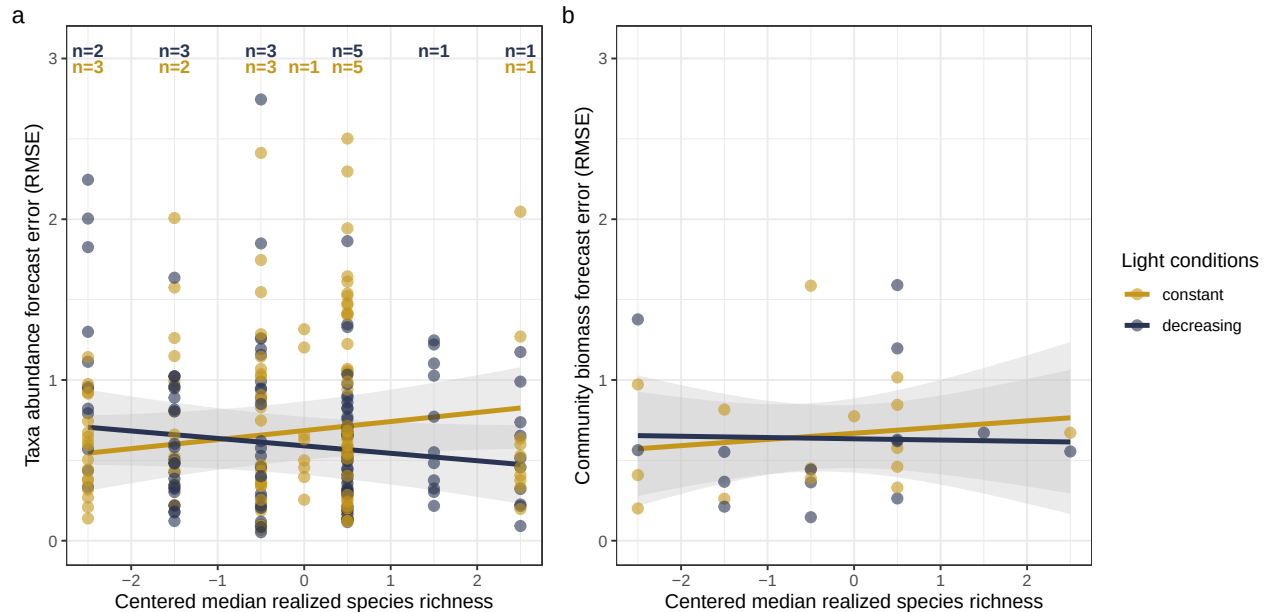


Figure 1: Effects of species richness and light on forecast error. The two treatments had interactive effects on forecast error of species abundances (a) but no effect on the forecast error of community biomass (b). Forecasts were computed with Simplex EDM, and forecast error was calculated as root mean square error (RMSE). Observed richness deviated from planned richness because not all species persisted throughout the experiment. We therefore used realized richness (i.e. the median of the observed richness values) as explanatory variable. Median richness was centered; a centered median richness of zero corresponds to a median richness of 5.5 (minimum median richness: 3; maximum median richness: 8). Lines display the fit of the linear mixed model; shaded areas denote the 95% confidence intervals. In (a), each point represents a

forecasted taxon in a bottle. Taxon, bottle, and incubator were included as random intercepts in the model. Inserts at the top indicate the number of bottles for the respective richness level (yellow: constant light, blue: decreasing light). In (b), bottle and incubator were included as random intercepts in the model.

2.3 Table

```
#### -----
#### Table for simplex species forecasts regression (Main text)
#### -----

conf.int.1 <- confint(m.simplex)
conf.int.1 <- conf.int.1[5:8, ]
conf.int.1 <- round(conf.int.1, 3)
conf.int.1 <- apply(conf.int.1, 1, paste, collapse = " - ")

spec.ab.tab <-
  data.frame(
    Response = "Forecast error\nof target\nabundances",
    Covariate = c(
      "Intercept",
      "Richness",
      "Light (decreasing)",
      "Richness:Light"
    ),
    Estimate = fixef(m.simplex),
    CI = conf.int.1,
    SS = c(NA, anova(m.simplex)[, 1]),
    DF = c(NA, paste(
      anova(m.simplex)[, 3], round(anova(m.simplex)[, 4], 1), sep = ", "
    )),
    F.value = c(NA, anova(m.simplex)[, 5]),
    p.value = c(NA, anova(m.simplex)[, 6])
  )

#### -----
#### Table for simplex community biomass regression (Main text)
```

```

#### -----

conf.int.2 <- confint(m.simplex.bio)
conf.int.2 <- conf.int.2[4:7, ]
conf.int.2 <- round(conf.int.2, 3)
conf.int.2 <- apply(conf.int.2, 1, paste, collapse = " - ")

bio.rmse.tab <-
  data.frame(
    Response = "Forecast error\nof community\nbiomass",
    Covariate = c(
      "Intercept",
      "Richness",
      "Light (decreasing)",
      "Richness:Light"
    ),
    Estimate = fixef(m.simplex.bio),
    CI = conf.int.2,
    SS = c(NA, anova(m.simplex.bio)[, 1]),
    DF = c(NA, paste(
      anova(m.simplex.bio)[, 3], round(anova(m.simplex.bio)[, 4], 1), sep = ", "
    )),
    F.value = c(NA, anova(m.simplex.bio)[, 5]),
    p.value = c(NA, anova(m.simplex.bio)[, 6])
  )

main.tab <- rbind(spec.ab.tab, bio.rmse.tab)

rownames(main.tab) <- NULL

main.tab$p.value <- ifelse(main.tab$p.value < 0.0001, "<0.0001",
  sprintf("%.4f", round(main.tab$p.value, 4)))

main.tab <- main.tab %>%
  dplyr::mutate(
    Estimate = round(Estimate, 3),

```

```

    SS = round(SS, 3),
    F.value = round(F.value, 3)
)

options(knitr.kable.NA = '')
main.tab %>%
  mutate_all(linebreak, align = "l") %>%
  kable(
    "latex",
    booktabs = T,
    digits = 3,
    col.names = c(
      "Response",
      "Covariate",
      "Estimate",
      "95\\% CI",
      "SS",
      "DF",
      "\\textit{F}-value",
      "\\textit{p}-value"
    ),
    caption = "Results of linear mixed models investigating the effects of
species richness and light on forecast errors of taxa and community biomass.
This table is an extended version of Table 1 in the main text and includes
both the regression tables and the analysis of variance (Type III) tables of
the models. The models included median centered richness, light treatment
and their interaction as fixed effects. The random intercepts were included
for the bottles, incubators and the targets (i.e. taxa). Forecast error is
based on forecasts computed with Simplex EDM.",
    align = c("llccrr"),
    escape = F,
    label = "MainTab"
) %>%
collapse_rows(
  columns = c(1),
  latex_hline = 'custom',

```

Table 1: Results of linear mixed models investigating the effects of species richness and light on forecast errors of taxa and community biomass. This table is an extended version of Table 1 in the main text and includes both the regression tables and the analysis of variance (Type III) tables of the models. The models included median centered richness, light treatment and their interaction as fixed effects. The random intercepts were included for the bottles, incubators and the targets (i.e. taxa). Forecast error is based on forecasts computed with Simplex EDM.

Response	Covariate	Estimate	95% CI	SS	DF	<i>F</i> -value	<i>p</i> -value
Forecast error of target abundances	Intercept	0.685	0.503 - 0.865				
	Richness	0.056	-0.008 - 0.121	0.008	1, 21.1	0.046	0.8328
	Light (decreasing)	-0.095	-0.265 - 0.081	0.225	1, 6.1	1.230	0.3092
	Richness:Light	-0.102	-0.193 - -0.014	0.922	1, 19.7	5.046	0.0363
Forecast error of community biomass	Intercept	0.668	0.456 - 0.879				
	Richness	0.039	-0.1 - 0.176	0.005	1, 19.5	0.060	0.8089
	Light (decreasing)	-0.034	-0.235 - 0.167	0.008	1, 13.1	0.108	0.7476
	Richness:Light	-0.046	-0.188 - 0.1	0.029	1, 13.4	0.387	0.5445

```

custom_latex_hline = c(1),
valign = "middle"
) %>%
kable_styling(font_size = 7)

```

3 Versions of R and of the R-packages used

R version: R version 4.3.2 (2023-10-31).

Base packages: stats, graphics, grDevices, utils, datasets, methods, base.

Other packages: kableExtra 1.4.0, lmerTest 3.1-3, lme4 1.1-35.1, Matrix 1.6-5, here 1.0.1, ggstance 0.3.6, patchwork 1.2.0, lubridate 1.9.3, forcats 1.0.0, stringr 1.5.1, dplyr 1.1.4, purrr 1.0.2, readr 2.1.5, tidyr 1.3.1, tibble 3.2.1, ggplot2 3.5.0, tidyverse 2.0.0, knitr 1.45.

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